Expansion of hybridoma

Feeder cells

• Feeder cells are necessary to encourage the growth of the hybridoma cells during

expansion from microwells, cloning and thawing from frozen storage.

• Three commonly used types of feeder cells are: thymocytes, splenocytes and peritoneal macrophages

• The source of the growth factors

Thymocytes:

• Thymus should be obtained from young animal (up to 10 weeks).

• It is not necessary to use the same species (Balb/c) as the hybridoma parents 2-1x100000000thymocytes/mouse should be obtained .

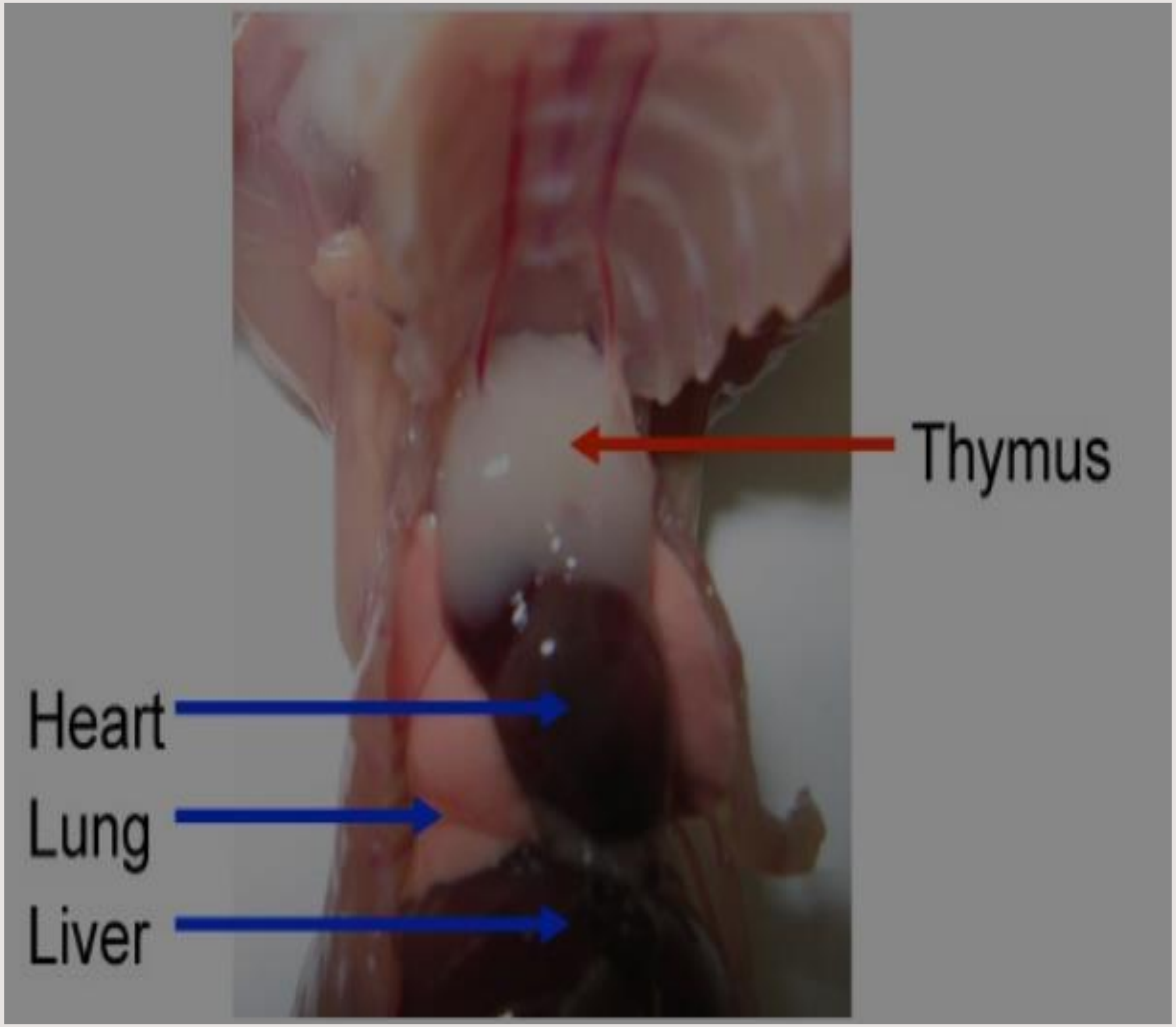
Preparation of thymic cells

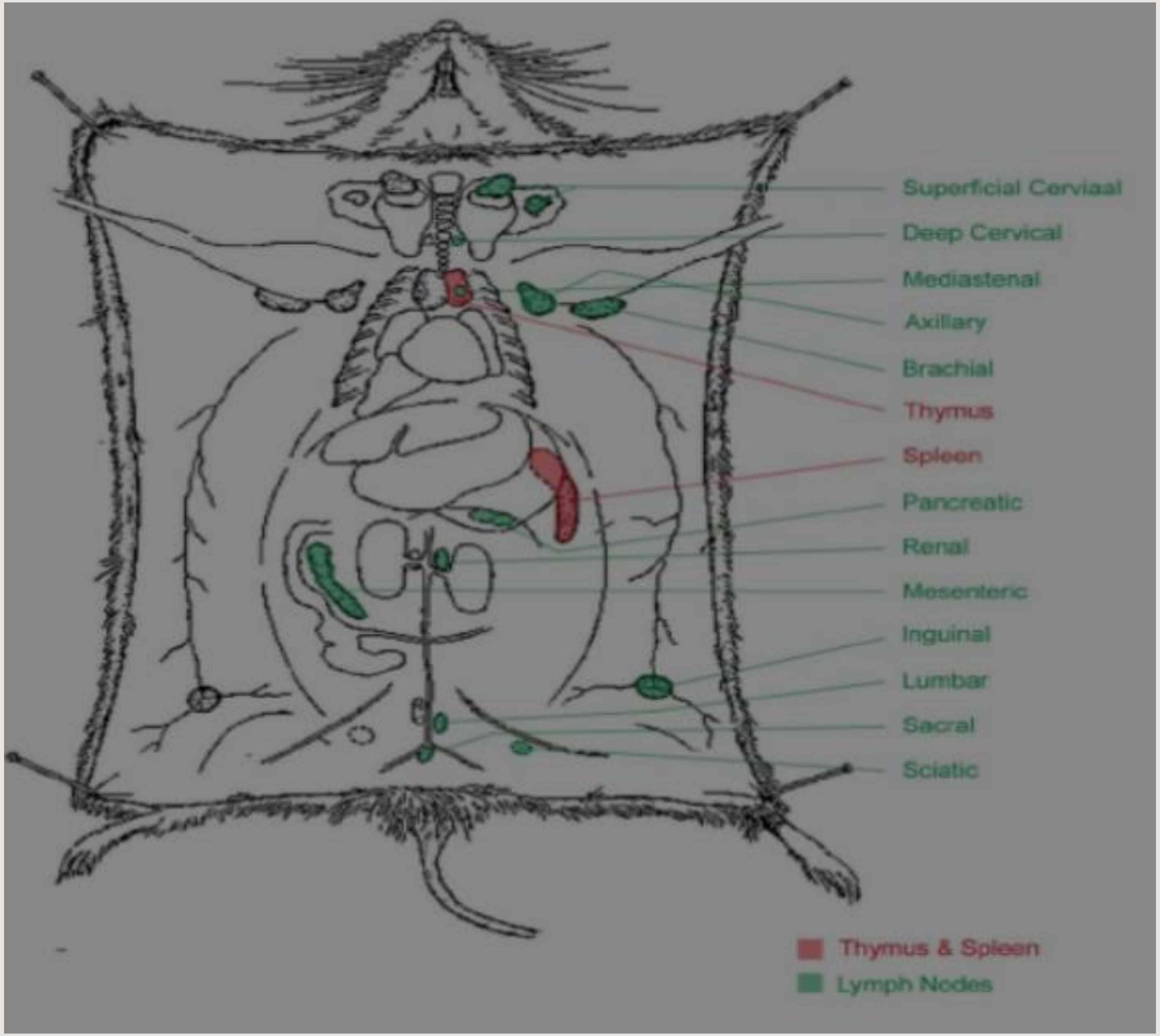
- Euthanize mouse in a CO_2 chamber. Put the animal on its back on a dissection mat, pin the feet on the mat and wet the fur by spraying with 70% ethanol.

- Make a midline incision through the skin, starting from above the urethral opening straight up to the lower jaw with scissors and extend the incision downward to the knees on both sides. The incision looks like an upside down “Y”. Pull the loose skin back on the sides, and pin it to the dissection mat.

- Puncture the diaphragm from the xiphoid, cut the ribs on each side up to about the clavicle, then lift up the ribs with forceps and pin it to the dissection mat.

- The thymus is just under the ribs, and looks like two thin white lobes overlying the heart (Figure 1). Disconnect connective tissue surrounding the thymus, gently pull and remove the pair of thymus lobes with curved serrated forceps. Place thymus lobes into a 6 well plate containing 5 ml RPMI-1640.


Thymus
Heart
Lung
Liver



Superficial Cerviaal
Deep Cervical
Mediastenal
Axillary
Brachial
Thymus
Spleen
Pancreatic
Renal
Mesenteric
Inguinal
Lumbar
Sacral
Sciatic
Thymus & Spleen
Lymph Nodes

Peritoneal Macrophages

- Swab with alcohol after anesthetization
- Remove the skin and in lower abdomen
- Inject ice-cold 5 ml PBS with 3% FCS using 27G needle
- Gently massage abdomen for 2 min.

- Collect fluid from peritonium using 25 G needle
- Spin and add 5 ml PBS
- Count the cells with trypan blue exclusion and haemocytometer: typically 5-10 million cells could be collected. 41% macrophages and granulocytes, 53% B cells, 6% T cells
- It is important not to make bleeding in the peritonium.